

$^{34}\text{S}(^3\text{He},\text{d})$ 2017Gi09,1969Gr23

$J^\pi=0^+$ for ^{34}S g.s.

- 2017Gi09:** a 20-MeV $^3\text{He}^{2+}$ beam was produced from the MP Van de Graaff tandem accelerator at the Maier-Leibnitz Laboratorium (MLL) and impinged on a $50\text{-}\mu\text{g}/\text{cm}^2$ Ag_2S target 99.999% enriched in ^{34}S with a $8\text{-}\mu\text{g}/\text{cm}^2$ carbon backing. Reaction products were momentum-separated by the Q3D magnetic spectrograph and focused on focal plane detectors consisting of two multiwire proportional counters (MWPCs) and a plastic scintillator. Deduced 22 levels, L-transfers, J, π , spectroscopic factors from FRESKO-DWBA analysis, and resonance strengths. Deduced the thermonuclear $^{34}\text{S}(\text{p},\gamma)^{35}\text{Cl}$ reaction rate through Monte Carlo methods using the STARLIB code. Calculated the $^{34}\text{S}/^{32}\text{S}$ isotopic ratio using the hydrodynamical nova model SHIVA.
- 1969Gr23:** a 11.0-MeV ^3He beam was produced from the MIT-ONR Van de Graaff generator. The target was $9.3\text{-}\mu\text{g}/\text{cm}^2$ ^{34}S made by evaporating PbS (67.6% ^{34}S) onto a formvar backing. Deuterons were detected on $50\text{ }\mu\text{m}$ nuclear emulsions in the MIT multiple-gap spectrograph at 23 different angles from 7.5° to 172.5° . Measured $\sigma(E_d, \theta)$. Deduced 17 levels, J, π , L-transfers, spectroscopic factors from zero-range JULIE-DWBA analysis.
- 1970Mo01:** a 13-MeV ^3He beam was produced from the Argonne tandem Van de Graaff accelerator. The target was $200\text{-}\mu\text{g}/\text{cm}^2$ 86% enriched ^{34}S . Reaction products were detected using two telescopes of silicon surface-barrier detectors with FWHM ≈ 80 keV and analyzed by a magnetic spectrograph with FWHM ≈ 20 keV at $\theta < 20^\circ$. Measured $\sigma(E_d, \theta)$, $\theta_{\text{c.m.}} \approx 5^\circ$ to 100° . Deduced levels, J, π , L-transfers, spectroscopic factors from local, zero-range JULIE-DWBA analysis.
- 1994Ve04:** a 25-MeV ^3He beam was produced from the Orsay MP Tandem Van de Graaff accelerator. The target was KCl with natural chlorine. Reaction products were momentum-analyzed by an Enge split-pole magnetic spectrograph and detected using a 50-cm long telescope of a position-sensitive proportional counter, proportional ΔE counter, and a plastic scintillator. Measured $\sigma(E_d, \theta)$. Deduced J, π , L-transfer, and spectroscopic factor for the ground state of ^{35}Cl from local, zero-range DWUCK-DWBA analysis.

 ^{35}Cl Levels

For levels below 6 MeV, spectroscopic factor $(2J+1)\text{C}^2\text{S} = \sigma(\theta)_{\text{exp}}/\sigma(\theta)_{\text{DWBA}}/N$, where N is a normalization factor. J is the spin of the populated ^{35}Cl level. $N=4.42$ is used in 1969Gr23 and 1970Mo01 from 1966Ba54. $N=4.43$ is used in 1994Ve04. For levels above 6 MeV, C^2S values are from 2017Gi09 without using the factor N. 2017Gi09 also states that it is not possible to distinguish between $J=L+1/2$ or $J=L-1/2$ coupling given the experimental uncertainties. Values of resonance strengths $\omega\gamma(\text{min})$ and $\omega\gamma(\text{max})$ are calculated by 2017Gi09 using their measured C^2S and calculated Γ_p , assuming the Γ_{tot} is dominated by Γ_γ and thus $\omega\gamma$ depends only on Γ_p .

E(level) [†]	L [‡]	Comments
0	2	S: $(2J+1)\text{C}^2\text{S}=2.88$ in 1969Gr23 for $J^\pi=3/2^+$; $\text{C}^2\text{S}=1.3$ in 1970Mo01 for $J^\pi=3/2^+$; $\text{C}^2\text{S}=1.07$ in 1994Ve04 for $J^\pi=3/2^+$.
1220 10	0	E(level): other: 1220 12 (1970Mo01). S: $(2J+1)\text{C}^2\text{S}=0.26$ in 1969Gr23 for $J^\pi=1/2^+$; $\text{C}^2\text{S}=0.28$ in 1970Mo01 for $J^\pi=1/2^+$.
1758 10		
2645 10		
2686 10	2	S: $(2J+1)\text{C}^2\text{S}=0.06$ in 1969Gr23 for $J^\pi=(3/2, 5/2)^+$.
3002 10	2	E(level): weighted average of 2997 10 (1969Gr23) and 3010 12 (1970Mo01). S: $(2J+1)\text{C}^2\text{S}=0.14$ in 1969Gr23 for $J^\pi=(5/2)^+$; $\text{C}^2\text{S}=0.04$ in 1970Mo01 for $J^\pi=5/2^+$.
3158 10	3	E(level): weighted average of 3156 10 (1969Gr23) and 3160 12 (1970Mo01). S: $(2J+1)\text{C}^2\text{S}=2.88$ in 1969Gr23 for $J^\pi=7/2^-$; $\text{C}^2\text{S}=0.66$ in 1970Mo01 for $J^\pi=7/2^-$.
3908 10		
3962 10	0	E(level): weighted average of 3956 10 (1969Gr23) and 3970 12 (1970Mo01). S: $(2J+1)\text{C}^2\text{S}=0.03$ in 1969Gr23 for $J^\pi=1/2^+$; $\text{C}^2\text{S}=0.06$ in 1970Mo01 for $J^\pi=1/2^+$.
4053 10	1	E(level): weighted average of 4048 10 (1969Gr23) and 4060 12 (1970Mo01). S: $(2J+1)\text{C}^2\text{S}=0.38$ in 1969Gr23 for $J^\pi=(3/2)^-$; $\text{C}^2\text{S}=0.15$ in 1970Mo01 for $J^\pi=(3/2)^-$.
4167 10	1	E(level): weighted average of 4165 10 (1969Gr23) and 4170 12 (1970Mo01). S: $(2J+1)\text{C}^2\text{S}=0.85$ in 1969Gr23 for $J^\pi=(3/2)^-$; $\text{C}^2\text{S}=0.28, 0.62$ in 1970Mo01 for $J^\pi=(3/2)^-$.
5010 15		
5163 15	(2)	S: $(2J+1)\text{C}^2\text{S}=0.11$ without specifying J (1969Gr23).
5409 12	1	E(level): weighted average of 5407 15 (1969Gr23) and 5410 12 (1970Mo01). L: 1 in 1969Gr23 and L=(1,2) in 1970Mo01.

Continued on next page (footnotes at end of table)

$^{34}\text{S}(^3\text{He},\text{d})$ **2017Gi09,1969Gr23 (continued)** ^{35}Cl Levels (continued)

E(level) [†]	L [‡]	Comments
5660 15	2	S: (2J+1)C ² S=0.07 in 1969Gr23 for $J^\pi=(3/2)^-$. E(level): 5670 multiplet reported in 1970Mo01. L: 2 in 1969Gr23, possibly a component of the 5670 multiplet with L=(2) in 1970Mo01.
5684 12	1	S: (2J+1)C ² S=0.47 in 1969Gr23 for $J^\pi=3/2^+$. E(level): 5670 multiplet reported in 1970Mo01. L: 1 in 1969Gr23, possibly a component of the 5670 multiplet with L=(2) in 1970Mo01.
5760 12	(0,1)	S: (2J+1)C ² S=0.11 in 1969Gr23 for $J^\pi=(3/2)^-$. E(level): weighted average of 5760 15 (1969Gr23) and 5760 12 (1970Mo01). L: 1 in 1969Gr23 and L=0 in 1970Mo01.
6284 4	(2)	S: (2J+1)C ² S=0.12 in 1969Gr23 for $J^\pi=(3/2)^-$; C ² S=0.21 in 1970Mo01 for $J^\pi=1/2^+$.
6329 4	(0,1)	S: (2J+1)C ² S=0.086 2.
6377 2	(2,3)	S: (2J+1)C ² S=0.0030 2 for L=0, 0.0044 4 for L=1.
6427 2	(3)	S: (2J+1)C ² S=0.0012 6 for L=2, 0.0024 8 for L=3.
6468 2	(1)	S: (2J+1)C ² S=0.049 2. $\omega\gamma(\text{min})=6.5\times10^{-24}$ eV 2, $\omega\gamma(\text{max})=2.2\times10^{-22}$ eV 1.
6491 2	(2)	S: (2J+1)C ² S=0.034 1. $\omega\gamma(\text{min})=4.2\times10^{-14}$ eV 5, $\omega\gamma(\text{max})=4.7\times10^{-14}$ eV 5.
6545 2	(0,1)	S: (2J+1)C ² S=0.072 2. $\omega\gamma(\text{min})=3.7\times10^{-13}$ eV 1, $\omega\gamma(\text{max})=4.1\times10^{-13}$ eV 1.
6643 2	(1)	S: (2J+1)C ² S=0.004 1 for L=0, 0.0028 4 for L=1. $\omega\gamma(\text{min})=7.4\times10^{-10}$ eV 13, $\omega\gamma(\text{max})=4.1\times10^{-9}$ eV 1.
6674 2	(1,2,3)	S: (2J+1)C ² S=0.0144 8. $\omega\gamma(\text{min})=8.5\times10^{-6}$ eV 10, $\omega\gamma(\text{max})=8.9\times10^{-6}$ eV 10.
6761 2	(0,1)	S: (2J+1)C ² S=0.0020 4 for L=1, 0.0048 6 for L=2, 0.008 1 for L=3. $\omega\gamma(\text{min})=2.4\times10^{-8}$ eV 2, $\omega\gamma(\text{max})=4.1\times10^{-6}$ eV 8.
6778 2	(1)	S: (2J+1)C ² S=0.0056 12 for L=0, 0.0032 4 for L=1. $\omega\gamma(\text{min})=2.4\times10^{-4}$ eV 3, $\omega\gamma(\text{max})=1.5\times10^{-3}$ eV 1.
6823 2	(1)	S: (2J+1)C ² S=0.0084 8. $\omega\gamma(\text{min})=9.4\times10^{-4}$ eV 12, $\omega\gamma(\text{max})=1.0\times10^{-3}$ eV 1.
6842 2	(2,3)	S: (2J+1)C ² S=0.006 4.
6866 2	(0,2)	S: (2J+1)C ² S=0.00216 6 for L=2, 0.035 2 for L=3. $\omega\gamma(\text{min})=2.7\times10^{-3}$ eV 3, $\omega\gamma(\text{max})=2.7\times10^{-3}$ eV 3.
7066 2	(1,2)	S: (2J+1)C ² S=0.0216 6 for L=2, 0.035 2 for L=3. $\omega\gamma(\text{min})=3.3\times10^{-5}$ eV 1, $\omega\gamma(\text{max})=7.0\times10^{-4}$ eV 3.
7103 2	(1,3)	S: (2J+1)C ² S=0.0160 2(L=0)+0.0660 6 (L=2). $\omega\gamma(\text{min})=3.0\times10^{-2}$ eV 3, $\omega\gamma(\text{max})=4.7\times10^{-2}$ eV 5.
7178 2	(2)	S: (2J+1)C ² S=0.0088 8 for L=1, 0.012 1 for L=2. $\omega\gamma(\text{min})=2.2\times10^{-2}$ eV 1, $\omega\gamma(\text{max})=2.2\times10^{-2}$ eV 1.
7194 2	(2)	S: (2J+1)C ² S=0.0184 12 for L=1, 0.025 2 for L=3. $\omega\gamma(\text{min})=6.0\times10^{-2}$ eV 3, $\omega\gamma(\text{max})=1.30$ eV 7.
7227 2	(0,1)	S: (2J+1)C ² S=0.032 3.
7273 2	(0,1)	S: (2J+1)C ² S=0.065 15 for L=0, 0.044 2 for L=1.
7361 2	(1)	S: (2J+1)C ² S=0.050 12 for L=0, 0.030 1 for L=1. $\omega\gamma(\text{min})=11$ eV 2, $\omega\gamma(\text{max})=34$ eV 3.
7398 2	(2,3)	S: (2J+1)C ² S=0.0040 8. $\omega\gamma(\text{min})=0.30$ eV 4, $\omega\gamma(\text{max})=3.0$ eV 6.
		S: (2J+1)C ² S=0.042 2 for L=2, 0.0656 24 for L=3. $\omega\gamma(\text{min})=0.18$ eV 1, $\omega\gamma(\text{max})=0.19$ eV 1.

[†] From 1969Gr23 for E(level)<6 MeV and from 2017Gi09 for E(level)>6 MeV, unless otherwise noted.

$^{34}\text{S}(^3\text{He,d})$ [2017Gi09,1969Gr23](#) (continued)

^{35}Cl Levels (continued)

[‡] From DWBA analysis of the measured $\sigma(\theta)$ in [1969Gr23](#) for $E(\text{level}) < 6$ MeV and in [2017Gi09](#) for $E(\text{level}) > 6$ MeV, unless otherwise noted. Firm assignments by [2017Gi09](#) are placed in parentheses by the evaluators because they are based on very few data points in a very narrow range of angles in the DWBA fit which may not yield definitive and firm results.